UNCERTAINTY-AWARE TRAFFIC ROUTING USING SEARCH-BASED ALGORITHMS

By Ali Ahmad, Atharv Kothekar, Akeem Shane

PROBLEM STATEMENT

- **Classical route planning** assumes **fixed travel times** and returns one deterministic shortest path.
- **Real traffic** is **uncertain** because congestion, incidents, and time-of-day patterns change edge costs.
- A route that is **optimal** on average **may still be unreliable** in practice.
- Our goal was to build an AI routing system that **balances efficiency with reliability** under **uncertain traffic**.

MAIN HYPOTHESIS

Search-based routing combined **with uncertainty** modeling will **produce routes** that are **more realistic** and **more dependable** than free-flow shortest-path planning alone.

DATASET

- Dataset includes **28 cities** and **41–42 road connections** stored in two CSV files.
- Each node stores **city, population, region**, and **GPS** coordinates.
- Each edge stores **distance, speed limit, road type,** and **free-flow** travel time.
- Three traffic regimes are modeled for every edge: **am_peak, off_peak, and pm_peak**.
- Congestion is represented with Gaussian multipliers:

$$\text{travel time} = \text{free-flow time} \times \mathcal{N}(\mu, \sigma)$$

- Samples are clamped to be at least 1.0 so traffic never makes travel faster than free flow.

PHASE 1: SYSTEM DESIGN

Classes:

- TrafficDistribution
- Edge
- Node
- TrafficGraph

Design Choices

- Real GPS coordinates for geographic realism.
- Haversine distance for the A* heuristic.
- Separate deterministic and probabilistic cost functions.

PHASE 2: DETERMINISTIC BASELINES

Uniform Cost Search

- Edge weight = free-flow travel time
- Expands least-cost frontier first
- Guarantees optimal deterministic path

A* Search

- Same edge cost as UCS
- Uses Haversine distance as heuristic
- Reduces unnecessary exploration while preserving optimality

Evaluation Goal

Compare path correctness and node expansions across representative route cases and edge cases.

PHASE 3: TIME-DEPENDENT TRAFFIC SEARCH

- Phase 3 **replaces free-flow edge weights with congestion-adjusted mean travel time.**
- Both UCS and A* accept a **time-period parameter**.
- Deterministic traffic cost:

$$c(e, t) = \text{free-flow time}(e) \times \mu_t(e)$$

- This allows the same source-goal pair to produce **different optimal routes at different times of day.**
- The design keeps the search framework unchanged and only swaps the cost function.

PHASE 4: MONTE CARLO ROUTING UNDER UNCERTAINTY

1. Sample one traffic scenario by drawing a travel time for every edge.
2. Run search on the sampled scenario.
3. Repeat for many trials.
4. Count how often each route is selected.
5. Re-simulate the chosen route to estimate reliability statistics.

Outputs

Most common path, route frequency, average nodes expanded, mean travel time, standard deviation, and threshold-risk estimates.

EVALUATION METHODS

- Three test classes per phase: **correctness** (exact path + cost), **behavioral invariants** (e.g. off-peak ≤ peak), **edge cases** (unreachable, parallel edges, source = goal).
- Metrics collected: **optimal path**, **total cost**, **nodes expanded**, and for Phase 4 the MC distribution (mean, std, min/max, threshold probability).
- Phase 4 is structurally different — output is a **route-choice distribution**, not a single path. Tests check **frequencies** and **reliability bands**, with **fixed seeds** for reproducibility.

RESULTS: DETERMINISTIC AND TRAFFIC-AWARE SEARCH

Route	Hops	UCS expanded	A* expanded	Reduction
Albany → Troy	1	2	2	—
Utica → Rome	1	2	2	—
Buffalo → Niagara Falls	1	3	2	−1
Binghamton → Elmira	1	3	2	−1
Buffalo → Utica	3	9	5	−4
NYC → Albany *(unreachable)*	—	10	10	—
Total		**30**	**24**	**−20%**

RESULTS: DETERMINISTIC AND TRAFFIC-AWARE SEARCH

Route	Off-peak	AM peak	PM peak	Observation
NYC → Long Island City	20.8 min	56.0 min	60.8 min	Same path; cost scales 1.3× → 3.5× → 3.8×.
Buffalo → Niagara Falls	17.0 (direct)	**28.1 (via Tonawanda)**	27.2 (direct)	**Path flips at AM only** — I-190 AM (1.7×) > PM (1.6×).

- Traffic **changes which route wins**, not just its cost — Buffalo → NF flips at AM only.
- **NYC Metro** shows **3× congestion** blowup; rural routes barely change.
- A* loses most of its edge under traffic — savings drop **from 20% to 8%**. Haversine is in miles, cost in inflated minutes, so the heuristic barely prunes.

RESULTS: PROBABILISTIC ROUTING

Monte Carlo route-choice report (1,000 trials, seed = 42, AM peak)

Route	Most-common path	Freq.	Mean (min)	Std	Min / Max	P(t > threshold)
Buffalo → Niagara Falls	**via Tonawanda** (I-290 + I-290/NY-31)	59.4%	28.09	1.25	23.7 / 32.3	6.5% > 30 min
NYC → Long Island City	via I-495	92.6%	56.05	5.62	36.0 / 74.1	24.5% > 60 min
NYC → Yonkers	via I-87	84.3%	56.06	7.02	31.0 / 78.6	29.3% > 60 min
NYC → Hempstead (3-hop)	via Long Island City	60.7%	148.47	12.99	104.4 / 189.6	—
Binghamton → Corning (off-peak)	via Elmira	100.0%	67.33	1.50	—	—

RESULTS: PROBABILISTIC ROUTING

- **Selection frequency isn't always a landslide** — Buffalo → NF picks Tonawanda only 59% of the time. Deterministic search hides this.
- **Reliability tracks where uncertainty lives** — rural std ≈ 1.5 min, NYC Metro std = 5–13 min.
- **Threshold probability enables "on-time" questions** — 1-in-4 chance NYC → LIC exceeds 60 min.

CONCLUSIONS AND INTERPRETATION

- **Uncertainty changes the question:** Deterministic search asks "how long will this take?"; probabilistic search asks "which route is most often optimal, and how reliable is it?"
- **Congestion variance matters most in cities:** High-mean edges amplify variance; rural routes are effectively deterministic.
- **MC outputs are operational**, not just academic — enables 95th-percentile travel time and "how likely am I to be late?" queries that single-path algorithms can't answer.

FUTURE WORKS

- **Increase** the **dataset size** to include more roads, intersections, and traffic patterns across different areas.
- **Add** more **traffic-contributing attributes**, such as weather, accidents, construction, time of day, events, and road conditions.
- **Expand testing with real-world routes** to improve accuracy and reliability across different traffic scenarios.

Scan QR code to see GitHub repo and detailed results

THANK YOU